

AWS Solutions Architect

Associate · SAA-C03 · Recall Poster

THE NUMBERS		pure rote — learn these first
S3 max object / single PUT	5 TB / 5 GB	
Min duration IA / Glacier / Deep	30 / 90 / 180 days	
S3 multipart part size	5 MB – 5 GB, max 10,000 parts	
Glacier expedited / standard / bulk	1–5 min / 3–5 h / 5–12 h	
Lambda max duration	15 min (900 s)	
Lambda memory	128 MB – 10,240 MB	
Lambda payload sync / async	6 MB / 256 KB	
Lambda default concurrency	1,000 per region	
SQS max message	256 KB	
SQS retention default / max	4 days / 14 days	
SQS visibility default / max	30 s / 12 h	
SQS long poll wait	20 seconds	
Kinesis shard in / out	1 MB/s / 2 MB/s	
Kinesis retention default / max	24 h / 365 days	
DynamoDB max item	400 KB	
DynamoDB RCU / WCU	4 KB strong read / 1 KB write	
RDS backup retention	7 days default, 0 – 35 range	
DynamoDB LSI / GSI per table	5 / 20	
Read replicas RDS / Aurora	5 / 15	
Max storage RDS / Aurora	64 TiB / 128 TiB	
VPC CIDR range	/16 down to /28	
Reserved IPs / usable in a /24	5 / 251	
Spot notice; spot & RI discount	2 min; ~90% / ~72%	
CloudWatch basic / detailed	5 min / 1 min	

EC2		compute
Instance states	pending, running, stopping, stopped, shutting-down, terminated, rebooting	
Survives reboot but not stop	instance store data + public IPv4	
Stop: lost / kept	RAM & instance store / EBS volumes	
Metadata endpoint, version to use	169.254.169.254, IMDSv2	
Hibernate saves	RAM, to the root EBS volume	
User data runs	once, at first boot, as root	
System check vs instance check	AWS problem, stop/start to move host — vs your OS or config	
Low latency / isolation / distributed	cluster / spread / partition placement group	
Socket or core-based licence	Dedicated Host	
Steady 24/7 vs interruptible batch	Savings Plan or RI vs Spot	
SSH user: Amazon Linux / Ubuntu	ec2-user / ubuntu (CentOS: centos, Debian: admin)	
AMI, ECR image, snapshot scope	region — must be copied elsewhere	

Trap: Dedicated **Instance** only isolates hardware per account. Only a Dedicated **Host** gives you the physical server and its socket and core counts.

AUTO SCALING & LOAD BALANCING	
Default cooldown and grace period	300 seconds each
Recommended scaling policy	target tracking
Known spike / learned pattern	scheduled / predictive scaling
Instances killed before app boots	raise the grace period; use an ELB health check
Group flapping in and out	raise cooldown, widen the alarm threshold
Script before termination / debug one	lifecycle hook / standby
Define the instances	launch template (configurations are legacy)
ALB / NLB / GWLB layer	7 / 4 / 3
Static IP, UDP, millions of req/s	NLB
Path or host routing, dynamic container ports, Cognito auth	ALB
Cross-zone default	ALB on, NLB off, Classic off
Min subnets, internet-facing	2 subnets in 2 AZs
Health check lives on / many certs	the target group / SNI

LAMBDA, CONTAINERS & APP PLATFORMS	
Needs longer than 15 minutes	Fargate, ECS, Batch or Step Functions
Faster with no code change	more memory (CPU scales with it)
Cold starts / starving the account	provisioned / reserved concurrency
429 TooManyRequests	concurrency limit hit
Share dependencies / failed async	layers / dead-letter queue
Lambda in a VPC needs internet	NAT gateway
ECS task role vs execution role	your code's permissions vs ECS pulling the image
Kubernetes or kubectl	EKS
Containers: no servers / needs GPU	Fargate / EC2 launch type
Upload code, AWS builds the stack	Elastic Beanstalk (Immutable or Blue/Green = zero downtime)
Chef or Puppet / long batch jobs	OpsWorks / AWS Batch
Edge: light rewrite / calls services	CloudFront Functions / Lambda@Edge

S3		object storage
Consistency model	strong read-after-write	
Default encryption today	SSE-S3, applied automatically	
SSE-S3 / SSE-KMS / SSE-C	AWS holds all keys / you own the CMK / you send the key every request	
Accidental vs permanent delete	versioning / MFA Delete (root + CLI)	
Undeletable for seven years	Object Lock, compliance mode	
Replication requires	versioning on both + an IAM role — and it never copies existing objects	
Slow uploads from far away	Transfer Acceleration	
Browser uploads one file securely	presigned URL	
CloudFront reads a private bucket	Origin Access Control (OAC)	
Stop all public access	S3 Block Public Access	
Hot / unknown pattern / infrequent	Standard / Intelligent-Tiering / Standard-IA	
Events can go to	SQS, SNS, Lambda, EventBridge	
Infrequent and re-creatable	One Zone-IA (single AZ)	
Archive: ms / minutes–hours / cheapest	Glacier Instant / Flexible / Deep Archive	

Trap: One Zone-IA lives in **one AZ**. If the question says "must survive the loss of an Availability Zone," it is wrong, however cheap it looks.

EBS, EFS, FSX & TRANSFER	
EBS scope	one AZ — never spans AZs
Move a volume to another AZ	snapshot, then create from it there
Encrypt an existing volume or RDS	snapshot → copy with encryption → restore
Changeable while running	type, size, IOPS (grow only, never shrink)
Snapshots stored in	S3, incremental, region-scoped
General default volume	gp3 (3,000 IOPS / 125 MB/s included)
Guaranteed IOPS / throughput / cold	io2 / st1 / sc1 (st1 & sc1 cannot boot)
Highest IOPS, data disposable	instance store
Shared files, Linux, many instances	EFS (NFS v4.1, regional)
SMB, Windows, Active Directory	FSx for Windows File Server
HPC or ML scratch, links to S3	FSx for Lustre
On-prem NFS/SMB backed by S3	S3 File Gateway
Shrink on-prem footprint vs keep it all local	Volume Gateway cached / stored mode
Retire a tape library	Tape Gateway
Petabytes, no bandwidth vs ongoing sync	Snowball / DataSync
One backup policy across services	AWS Backup

DATABASES	
Multi-AZ purpose	availability — synchronous, not readable
Read replica purpose	read scaling — async, readable, cross-region
Multi-AZ failover	automatic, 60–120 s, same endpoint
RDS gives you OS access	no — need it? run on EC2
Aurora copies / failover	6 across 3 AZs / under 30 s
Restoring a snapshot gives	a new instance with a new endpoint
Cross-region DR, sub-second lag	Aurora Global Database
Intermittent load, near-zero idle cost	Aurora Serverless v2
DynamoDB microsecond reads	DAX (read-through only)
LSI vs GSI	same PK, made with the table / new PK any time, eventually consistent
Active-active across regions	Global Tables
Every item change / expire free	DynamoDB Streams + Lambda / TTL
ProvisionedThroughputExceeded	hot partition → better key, or on-demand
Redis vs Memcached	persistence, failover, sorted sets / simple multi-threaded cache

Warehouse vs ad-hoc SQL on S3	Redshift / Athena
Short queries stuck behind long	Redshift WLM query queues
Graph / time series / ledger	Neptune / Timestream / QLDB
Migrate engine, low downtime	DMS + Schema Conversion Tool

Trap: you can never read from a Multi-AZ standby. If an answer offers the standby for reporting queries it is wrong — that is what a read replica is for.

VPC & CONNECTIVITY	
What makes a subnet public	a route to an internet gateway
Private instances reach the internet	NAT gateway in a public subnet (one per AZ)
Security group	instance level, stateful, allow-only
Network ACL	subnet level, stateless, allow + deny, in rule order
Block a single malicious IP	NACL deny rule (groups cannot deny)
Default vs custom NACL	allows everything / denies everything
Ephemeral port range	1024 – 65535
Two VPCs vs hundreds	peering (not transitive) / Transit Gateway
Expose your service to another VPC	PrivateLink + NLB
Gateway endpoints work with	S3 and DynamoDB only (free)
Interface endpoint	everything else — an ENI with a security group
Hybrid in days vs steady bandwidth	Site-to-Site VPN / Direct Connect (weeks)
Why traffic is being dropped	VPC Flow Logs — look for REJECT
Shell access, no open ports, audited	SSM Session Manager

Trap: NACLs are stateless. An inbound rule alone is never enough — you must also allow the **outbound ephemeral ports** so the reply can get out.

ROUTE 53 & THE EDGE	
A/B test, gradual migration	Weighted
Active-passive disaster recovery	Failover
Compliance, language, licensing	Geolocation (the user's location)
Shift load between regions with a bias	Geoproximity (your resources)
Fastest region for each user	Latency
Up to 8 healthy records	Multivalue answer
Works at the zone apex, free	Alias (a CNAME cannot)
CloudFront default / max TTL	24 hours / 1 year
Health check failure threshold	3 consecutive checks
CloudFront certificate must live in	us-east-1
Restrict content to paying users	signed URLs or signed cookies
Static anycast IPs, TCP/UDP, gaming	Global Accelerator (does not cache)

SECURITY & IDENTITY		30% of the exam
Policy evaluation order	explicit Deny → explicit Allow → implicit deny	
Never grant, only limit	SCPs, permission boundaries, session policies	
SCP effect on the management account	none — it is never restricted	
App on EC2 needs S3	IAM role via an instance profile	
Another account needs access	cross-account role + sts:AssumeRole	
Workforce sign-in / mobile users	IAM Identity Center / Cognito identity pools	
Resource-based policy must have	a Principal	
KMS direct encryption limit	4 KB → above that, envelope encryption	
Single-tenant, FIPS 140-2 Level 3	CloudHSM	
Rotate DB creds / free config store	Secrets Manager / SSM Parameter Store	
SQL injection, XSS, rate limit by IP	AWS WAF	
DDoS Response Team, cost protection	Shield Advanced	
GuardDuty / Inspector / Macie	detects threats / scans for CVEs / classifies PII	
Detective / Security Hub	investigates / collects all findings	
Trusted Advisor categories	Cost, Performance, Fault tolerance, Security, Service limits	
Four tag categories	Technical, Automation, Business, Security	

INTEGRATION & ANALYTICS	
Decouple a spiky front end	SQS + auto-scale workers on queue depth
Standard queue guarantees	at-least-once, best-effort order
Exactly-once and strict order	FIFO queue (name ends .fifo)
Message processed twice	visibility timeout shorter than processing
Poison messages / empty receives	dead-letter queue / long polling
One message to many queues	SNS fan-out (+ filter policies)
Filter on event body, SaaS, replay	EventBridge
Coordinate steps, retries, approvals	Step Functions (SWF is legacy)
App already speaks AMQP, MQTT, JMS	Amazon MQ
Many consumers read the same records	Kinesis Data Streams
Load a stream to S3 or Redshift, no code	Firehose (~60 s, not real time)
Ad-hoc SQL on S3 / cut that bill	Athena / Parquet, compress, partition
Hadoop & Spark / search / BI	EMR / OpenSearch / QuickSight

How to use this			
Left of each row is the cue , right is the answer . Lay a ruler down the middle of a column, work down saying answers out loud, then slide it away to check. Blocks are grouped by area, so you can drill one area at a time. Amber strips are traps — where a plausible-looking answer is wrong.			
30% Secure	26% Resilient	24% High-performing	20% Cost-optimized

MONITORING, GOVERNANCE & COST	
CloudWatch / CloudTrail / Config	performance / who did what / configuration and compliance
Metrics needing the agent	memory and disk space used
Count ERROR lines and alarm	metric filter on a log group
CloudTrail free by default	90 days of management events
Who deleted an S3 object	enable data events
Prove logs untampered / all regions	log file integrity validation / apply the trail everywhere
Fix a non-compliant resource	Config remediation via SSM Automation
Only required CFN section / preview	Resources / change set
Keep data on delete / spot drift	DeletionPolicy Retain / drift detection
Many accounts and regions / guard rails	StackSets / Control Tower + SCPs
Right-size / alert on spend	Compute Optimizer / AWS Budgets
Cut the NAT bill for S3 traffic	S3 gateway endpoint
Survive an AZ vs a region failure	multiple AZs / multiple regions

SCENARIO TRIGGERS		highest marks per minute
"minimal operational overhead"	the serverless or managed option	
"highly available"	multiple AZs	
"disaster recovery", "region outage"	multiple regions	
"cannot be interrupted"	Spot is wrong, however cheap	
"decouple" / "notify many subscribers"	SQS / SNS	
"microseconds" / "sub-millisecond"	DAX / ElastiCache	
"static IP for a firewall allow-list"	NLB or Global Accelerator	
"Windows file share"	FSx for Windows, never EFS	
"no bandwidth, petabytes, deadline"	Snow family	
"urgent" vs "consistent bandwidth"	Site-to-Site VPN / Direct Connect	
"do not open port 22"	Session Manager	
"never hardcode credentials"	an IAM role	
"must be the only key holders"	CloudHSM, or SSE-C for S3	
"longer than 15 minutes"	not Lambda — Fargate, Batch, Step Functions	
"we already use Kubernetes"	EKS	
"must not leave the AWS network"	a VPC endpoint	
"pay per query, nothing running"	Athena	
"petabyte warehouse, high concurrency"	Redshift	
"MongoDB compatible" / "Cassandra"	DocumentDB / Keyspaces	
"no central trusted authority"	Managed Blockchain	
"single pane of glass" for findings	Security Hub	
"socket or core-based licence"	Dedicated Host	
TIE-BREAKERS, IN ORDER		

RESILIENCE & DISASTER RECOVERY	
RTO / RPO	how long you may be down / how much data you may lose
DR strategies, cheapest first	backup and restore, pilot light, warm standby, multi-site active-active
Backup and restore	RTO in hours — only backups in the other region
Pilot light	tens of minutes — data replicating, servers off
Warm standby	minutes — a scaled-down but working copy
Multi-site active-active	near zero — full capacity, both regions serving
Stateless tier survives instance loss	Auto Scaling across AZs behind an ELB
Database survives an AZ loss	Multi-AZ
Database survives a region loss	cross-region replica or Aurora Global Database
Objects survive a region loss	S3 Cross-Region Replication
Static page during an outage	Route 53 failover to an S3 website
Guard against accidental deletion	versioning + MFA Delete + Object Lock

COST OPTIMIZATION		20% of the exam
Steady 24/7 workload	Savings Plans or Reserved Instances	
Fault-tolerant, interruptible work	Spot Instances	
Idle at night and weekends	scheduled scaling or an instance scheduler	
Over-sized instances	Compute Optimizer	
Big NAT bill for S3 traffic	an S3 gateway endpoint	
Everything sitting in S3 Standard	lifecycle rules + Intelligent-Tiering	
Old snapshots, unattached volumes	find with Trusted Advisor, automate with DLM	
Incomplete multipart uploads	lifecycle rule to abort them	
High Athena bill	Parquet, compression, partitioning	
Spiky DynamoDB traffic	on-demand capacity mode	
Cross-AZ data transfer charges	keep chatty tiers in one AZ	
Warn before overspending	AWS Budgets + Cost Anomaly Detection	
See cost by team	cost allocation tags + Cost Explorer	

WHAT IS GLOBAL, REGIONAL, OR ZONAL	
Global services	IAM, Route 53, CloudFront, Organizations, Billing
Regional	EC2, S3 buckets, RDS, VPC, DynamoDB tables
AZ-scoped	subnets, EC2 instances, EBS volumes, ENIs, NAT gateways
Must be copied cross-region	AMIs, EBS & RDS snapshots, ECR images, KMS keys
A region has	3 or more AZs, independent power, cooling and network
Local Zone / Wavelength / Outposts	near a big city / in a 5G network / in your datacentre
Edge locations	CloudFront points of presence — far more numerous than regions

MNEMONICS	
Six pillars — Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.	
Trusted Advisor: CPFSS — Cost, Performance, Fault tolerance, Security, Service limits.	
DR, cheapest first — Backup and restore, Pilot light, Warm standby, Multi-site active-active.	
RTO = Time you may be down. RPO = Point in the past you fall back to.	
Placement groups — Cluster = Close, Spread = Separate, Partition = Pods.	
Default VPC gives five — internet gateway, main route table, security group, network ACL, DHCP option set.	
Groups guard instances. Lists guard subnets. Default NACL is open; a custom NACL is closed.	
Multi-AZ is durability. Read replicas are scalability.	
Geolocation = the user. Geoproximity = your resources.	
Streams if I write the consumer. Firehose if AWS writes it.	
GuardDuty detects. Inspector scans. Macie classifies. Detective investigates. Security Hub collects.	
CloudWatch watches performance. CloudTrail trails API calls. Config tracks configuration.	
Beanstalk deploys the app. CloudFormation declares the infrastructure. OpsWorks is Chef and Puppet.	